

SHOPNOW — BUSINESS ANALYSIS **CASE STUDY**

Sales Performance & Customer Analytics Dashboard

Tools Used: PostgreSQL, SQL, Power BI, DAX

Project Type: Business Analysis & Data Analytics

Dataset: ShopNow E-commerce Sales Dataset

1. Executive Summary

ShopNow is an e-commerce business that needs a clear understanding of its sales performance, customers, products, and geographic markets.

This Business Analysis project analyzes ShopNow's sales data using PostgreSQL and Power BI. SQL was used to investigate sales, customer behavior, product performance, and city-level performance, while Power BI was used to transform the analysis into an interactive management dashboard.

The final dashboard provides management with a consolidated view of key performance indicators, product performance, customer contribution, order activity, and business opportunities.

The analysis identified several important findings, including Furniture as the strongest-performing category, Peshawar as the highest-performing city, Usman as the highest-value customer, and Office Chair as the highest-revenue product.

2. Business Problem

ShopNow's management needs better visibility into its sales performance.

Without a centralized analytical dashboard, it can be difficult to determine:

- Which product categories generate the most sales?
- Which products contribute the most revenue?
- Which customers generate the highest sales?
- Which cities perform best?
- How many orders are being generated?
- What is the average order value?
- Where should management focus its sales and marketing efforts?

The objective of this project was therefore to analyze the available sales data and create a management dashboard that converts raw data into actionable business insights.

3. Business Requirements

The following business requirements were identified:

❖ Sales Performance

- Calculate total sales.
- Calculate total number of orders.
- Calculate average order value.
- Calculate total quantity sold.

❖ Product Analysis

- Identify the highest-performing product categories.
- Identify products generating the highest revenue.
- Identify products with the highest quantity sold.

❖ Customer Analysis

- Identify high-value customers.
- Analyze orders by customer.
- Compare customer sales performance.
- Analyze average order value by customer.

❖ Geographic Analysis

- Identify the highest-performing cities.
- Compare sales performance across cities.

❖ Management Reporting

- Develop an interactive Power BI dashboard.
- Present key findings through visualizations.
- Provide business recommendations based on the analysis.

4. Key Business Questions

The analysis was designed to answer the following questions:

1. What is ShopNow's total sales?
2. How many orders have been placed?

3. What is the average order value?
4. How many units have been sold?
5. Which product category generates the most sales?
6. Which product generates the highest revenue?
7. Which product has the highest quantity sold?
8. Which customer generates the most sales?
9. Which city generates the most sales?
10. What actions should management take based on the findings?

5. Data Sources

The project uses a relational ShopNow sales database created in PostgreSQL.

The main tables are:

❖ Orders

Contains order-level information such as:

- Order ID
- Customer ID
- Order Amount

❖ Order Items

Contains individual product-level order information:

- Order Item ID
- Order ID
- Product ID
- Quantity

❖ Products

Contains product information:

- Product ID
- Product Name
- Category
- Price

❖ Customers

Contains customer information:

- Customer ID
- Customer Name
- City
- Age

6. Data Model

The database follows a relational structure.

The key relationships are:

❖ **Customers → Orders**

Customers are connected to orders through `customer_id`.

❖ **Orders → Order Items**

Orders are connected to order items through `order_id`.

❖ **Products → Order Items**

Products are connected to order items through `product_id`.

This structure allows customer, order, and product information to be analyzed together.

The same relationships were implemented in Power BI to support interactive analysis.

7. Data Analysis Using SQL

PostgreSQL was used to perform the initial business analysis.

SQL was used to calculate:

- Sales by city
- Product quantity sold
- Orders by customer
- Customer sales and average order value
- Product-level revenue

For example, sales by city was analyzed by joining the Orders and Customers tables and grouping sales by city.

Product performance was analyzed by joining Order Items with Products and calculating the quantity sold and product revenue.

Customer performance was analyzed by calculating total sales, total orders, and average order value for each customer.

8. Power BI Dashboard

The SQL analysis was then connected to Power BI to create an interactive management dashboard.

The dashboard contains three pages.

➤ Page 1 — Executive Sales Overview

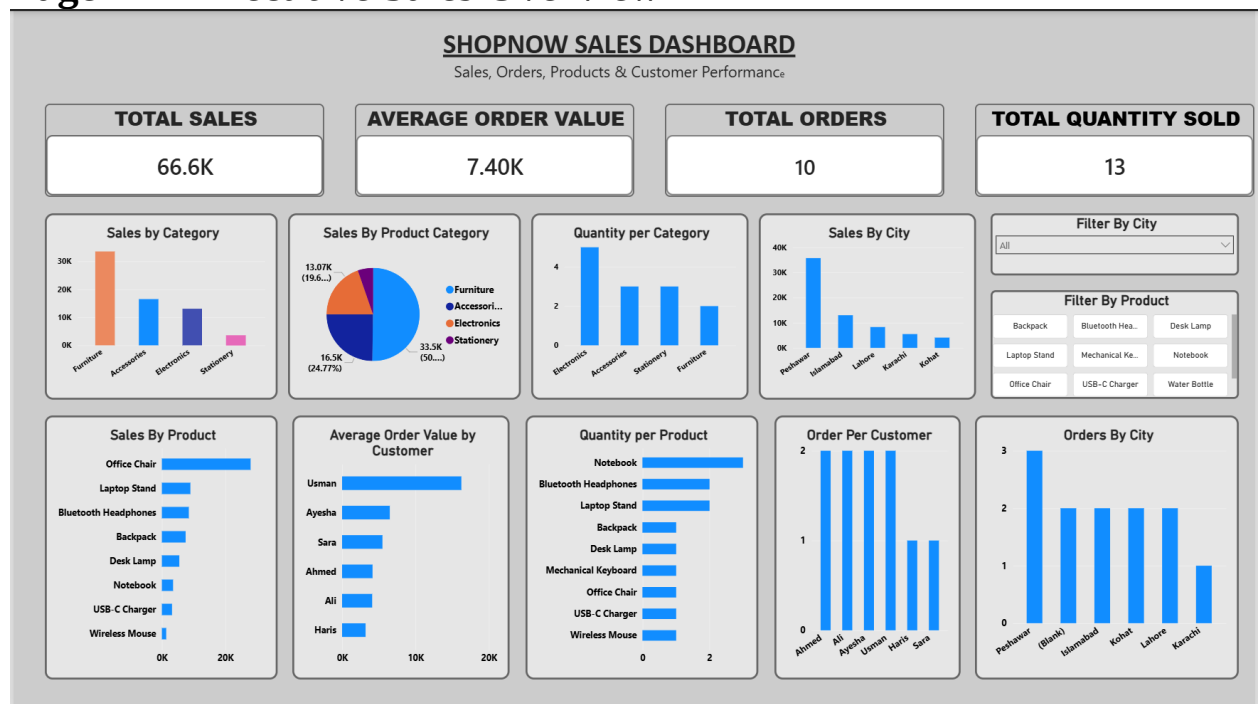


Figure 1: Executive Sales Overview

This page provides management with a high-level overview of ShopNow's performance.

Key KPIs include:

- **Total Sales:** 66.6K
- **Total Orders:** 10
- **Average Order Value:** 7.4K
- **Total Quantity Sold:** 13

The page also contains visualizations for:

- Sales by Product Category
- Sales by City

- Customer performance
- Product performance
- Orders by customer
- Product quantity
- Interactive city and product filters

This page answers the question:

"What is happening in the business?"

➤ Page 2 -Product & Customer Analysis



Figure 2: Product & Customer Analysis

The second page focuses on understanding the products and customers that drive ShopNow's sales performance. While the Executive Sales Overview provides a high-level view of the business, this page provides a deeper analysis of individual products and customer behavior.

The page contains five visualizations covering product sales, quantity sold, customer sales, orders by customer, and average order value by customer.

The **Sales by Product** visualization identifies which products generate the highest revenue, while **Quantity Sold by Product** highlights products with stronger unit demand. This distinction is important because a product with high sales volume is not necessarily the product generating the highest revenue.

The customer analysis allows management to identify high-value customers and compare their contribution to overall sales. **Sales by Customer** highlights the customers generating the most revenue, while **Orders by Customer** shows purchasing activity. **Average Order Value by Customer** provides additional insight into the value of individual customer transactions.

This page helps answer the key business question:

"Which products and customers are driving ShopNow's performance?"

The analysis identified **Office Chair** as the highest-revenue product at approximately **28K**, while **Usman** was the highest-value customer with approximately **32.5K** in sales.

➤ **Page 3 — Business Insights & Recommendations**



Figure 3: Business Insights & Recommendations

The third page converts the analytical findings into actionable business insights and recommendations. Its purpose is to move beyond reporting numbers and help management understand what actions can be taken based on the analysis.

The page summarizes five major findings:

- **Furniture** is the strongest-performing category, generating approximately **33.5K** in sales.
- **Peshawar** is the highest-performing city, generating approximately **35.7K** in sales.
- **Usman** is the highest-value customer, contributing approximately **32.5K** in sales.

- **Office Chair** is the highest-revenue product, generating approximately **28K** in product revenue.
- **Notebook** has the highest quantity sold, with **3 units**.

Based on these findings, the dashboard recommends focusing on high-performing categories, promoting high-revenue products, strengthening the Peshawar market, retaining high-value customers, and increasing spending from lower-value customers through cross-selling and product bundles.

This page demonstrates the Business Analyst's role in translating **data into decisions**.

It answers the final management question:

"What should the business do based on the analysis?"

9. Product & Customer Analysis

The second dashboard page provides more detailed analysis of products and customers.

The page contains five charts covering:

- Sales by Product
- Quantity Sold by Product
- Sales by Customer
- Orders by Customer
- Average Order Value by Customer

This page allows management to identify the products and customers that contribute most significantly to business performance.

It answers the question:

"Which products and customers are driving the business?"

10. Business Insights & Recommendations

The third dashboard page summarizes the major findings and converts them into business recommendations.

➤ **Key Insight 1 — Top Category**

- **Furniture generated approximately 33.5K in sales.**

Furniture is the strongest-performing product category in the analysis.

Recommendation

ShopNow should maintain strong inventory availability for Furniture and consider targeted promotions and bundles within this category.

➤ **Key Insight 2 — Top City**

- **Peshawar generated approximately 35.7K in sales.**

Peshawar is the strongest-performing city in the dataset.

Recommendation

ShopNow should strengthen customer retention and marketing activities in Peshawar while investigating what factors are contributing to its strong performance.

➤ **Key Insight 3 — Top Customer**

- **Usman generated approximately 32.5K in sales across 2 orders.**

This makes Usman the highest-value customer in the analyzed dataset.

Recommendation

ShopNow should focus on retaining high-value customers through loyalty programs, personalized offers, and targeted promotions.

➤ **Key Insight 4 — Top Product by Revenue**

- **Office Chair generated approximately 28K in product revenue.**

The Office Chair is the strongest product by revenue in the product-level analysis.

Recommendation

ShopNow should consider promoting the Office Chair through targeted campaigns, product bundles, and cross-selling opportunities.

➤ **Key Insight 5 — Best-Selling Product by Quantity**

- **Notebook recorded the highest quantity sold with 3 units.**

Although quantity sold and revenue are different performance measures, Notebook demonstrates strong unit demand within the available dataset.

Recommendation

ShopNow can use high-volume products such as Notebook as potential cross-selling or bundle products alongside higher-value items.

11. Business Recommendations

Based on the complete analysis, the following actions are recommended:

1. Prioritize high-performing categories

Furniture should receive continued attention because it is the strongest category by sales.

2. Promote high-revenue products

Office Chair should be included in targeted promotions and cross-selling campaigns.

3. Strengthen the Peshawar market

Since Peshawar generates the highest sales, ShopNow should investigate successful customer segments and marketing strategies in this market.

4. Retain high-value customers

Customers such as Usman should receive loyalty incentives and personalized offers to encourage repeat purchases.

5. Increase spending from lower-value customers

ShopNow can use product bundles, cross-selling, and personalized recommendations to increase customer spending.

6. Monitor product demand

Management should regularly compare revenue and quantity sold because a product with high unit sales is not necessarily the product generating the most revenue.

12. Limitations of the Analysis

There are some limitations in the available dataset.

No Order Date

The dataset does not contain an order date column.

Therefore, the project does not currently provide:

- Monthly sales trends
- Year-over-year growth

- Seasonal analysis
- Monthly customer acquisition
- Time-based forecasting

No Cost Data

The dataset does not contain product cost information.

Therefore, actual profit or profit margin cannot be calculated.

The correct profitability calculation would require:

$$\text{Profit} = (\text{Selling Price} - \text{Cost Price}) \times \text{Quantity Sold}$$

Future analysis should therefore include product cost data.

Small Dataset

The current dataset contains only a limited number of orders and customers.

Therefore, the findings demonstrate the analytical approach but should not be treated as representative of a large real-world e-commerce operation.

13. Tools & Technologies

➤ PostgreSQL

Used for:

- Database management
- Data exploration
- SQL queries
- Business analysis

➤ SQL

Used for:

- Aggregation
- Joins
- Grouping
- Customer analysis
- Product analysis
- Geographic analysis

➤ Power BI

Used for:

- Data visualization
- KPI reporting
- Interactive dashboards
- Business storytelling

➤ DAX

Used for:

- Calculated measures
- Product sales calculations
- Category-level analysis
- KPI calculations

14. Business Analysis Workflow

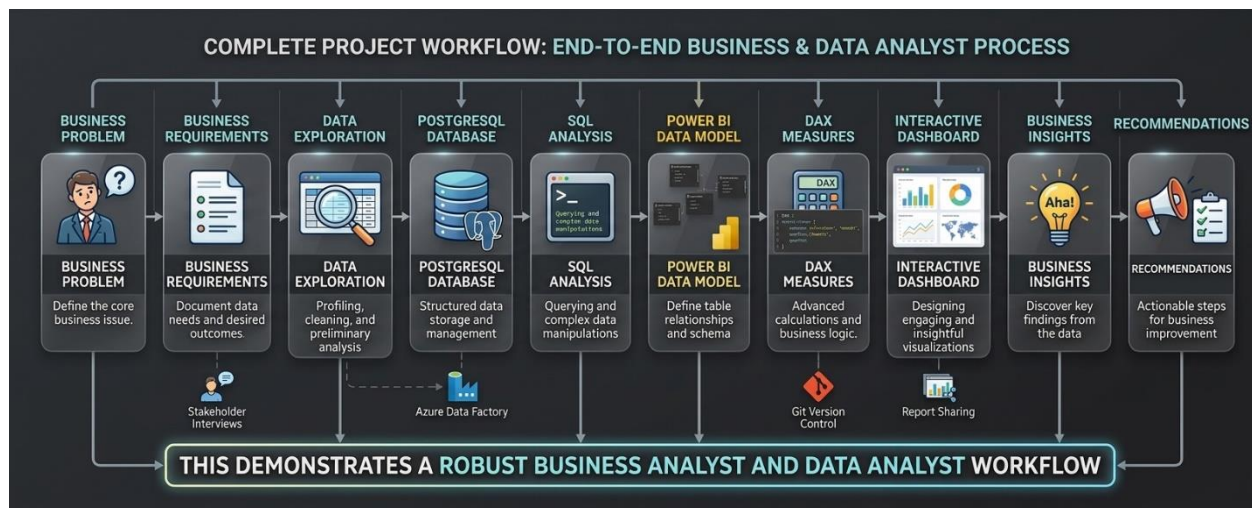


Figure 4: Analytics Project Life Cycle: From Problem to Business Impact

The complete project followed this workflow:

Business Problem



Business Requirements



Data Exploration



PostgreSQL Database



SQL Analysis



Power BI Data Model



DAX Measures



Interactive Dashboard



Business Insights



Recommendations

This demonstrates an end-to-end Business Analyst and Data Analyst workflow.

15. Final Conclusion

The ShopNow Business Analysis project demonstrates how raw transactional data can be transformed into meaningful business intelligence.

The analysis identified the strongest categories, products, customers, and geographic markets and presented these findings through an interactive Power BI management dashboard.

The most important findings were:

- **Total Sales:** 66.6K
- **Total Orders:** 10

- **Average Order Value:** 7.4K
- **Total Quantity Sold:** 13
- **Top Category:** Furniture — 33.5K
- **Top City:** Peshawar — 35.7K
- **Top Customer:** Usman — 32.5K
- **Top Product by Revenue:** Office Chair — 28K
- **Best-Selling Product by Quantity:** Notebook — 3 units

The project provides management with a structured way to understand current performance and identify opportunities for improving sales, customer retention, product promotion, and market focus.

16. Portfolio Project Summary

1. ShopNow Sales & Customer Analytics Dashboard

Developed an end-to-end Business Analysis project using PostgreSQL, SQL, Power BI, and DAX. Analyzed sales, customers, products, orders, and geographic performance; developed an interactive three-page management dashboard; identified key revenue drivers; and translated analytical findings into actionable business recommendations.

2. Key Skills Demonstrated:

SQL | PostgreSQL | Power BI | DAX | Data Analysis | Business Analysis | Data Visualization | KPI Reporting | Customer Analysis | Product Analysis | Business Insights